Technical Appendix

Potemkin Findings: Calibrating AI Research Loops with Null Worlds

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1 Formal framework and validity of loop-level inference

1.1 Objects

Let firms $i = 1, \ldots, N$ be partitioned into strata $g(i)$. The data are $D = (Y, E, M, X, g)$:

- Y is the panel of outcomes;
- $E = (E_1, \ldots, E_N)$ is the firm-level treatment (possibly several proxies per firm);
- M is the firm-level moderators;
- X is the panel of controls.

A *design space* S is a finite set of specifications. Each specification is a tuple (outcome proxy, treatment proxy, control set, fixed-effect structure, sample rule, transformation, clustering, moderator subset).

A *search policy* π is a deterministic rule that:

- visits specifications in an order that may depend on results already seen;
- stops by a rule that may also depend on those results;
- returns a report R.

The *loop* is the map $L_{S,\pi} : D \mapsto R$, and $T(R)$ is a real-valued evidence statistic of the report. In this paper T is either the largest sign-consistent H1 t-statistic visited or the smallest sign-consistent t-statistic among the four hypotheses in the reported package.

For a permutation σ of firm labels, $D^\sigma = (Y, E \circ \sigma, M, X, g)$ reassigns each firm the treatment of firm $\sigma(i)$ and leaves everything else unchanged. Let $\mathcal{G}$ be the group of permutations that map each stratum onto itself.

Assumption 1 (Within-stratum exchangeability under the null)**.** *Under H_0, $D^\sigma \overset{d}{=} D$ for every $\sigma \in \mathcal{G}$.*

Assumption 1 holds, for example, when outcomes, moderators, and controls do not depend on the treatment (the sharp null) and the treatment is assigned exchangeably within strata. In observational data it is a design-based assumption. It is stronger than the conditional parallel-trends assumption behind the difference-in-differences estimates in Section 5. Null-World Calibration therefore prices *search*; it does not by itself establish causality.

Theorem 1 (Validity of loop-level randomization inference)**.** *Let L be any deterministic loop whose design space S and policy π are fixed independently of the realized data, and let T be any real-valued statistic of its report. Draw $\sigma_1, \ldots, \sigma_B$ independently and uniformly from $\mathcal{G}$, independently of D. Define*

$$p_L = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{T(L(D^{\sigma_b})) \geq T(L(D))\}}{B+1}.$$

Under Assumption 1, $\Pr(p_L \leq \alpha) \leq \alpha$ *for every* $\alpha \in (0,1)$ *and every* $B \geq 1$.

Proof. Write $V(D) = T(L(D))$. Because L and T are fixed functions, V is a fixed measurable function of the data.

Step 1: the $B+1$ datasets are exchangeable. Let ρ be uniform on $\mathscr{G}$, independent of D and of (σ_b). By Assumption 1, $D \overset{d}{=} D^{\rho}$. Applying the map $D \mapsto (D, D^{\sigma_1}, \ldots, D^{\sigma_B})$ to both sides gives

$$(D, D^{\sigma_1}, \ldots, D^{\sigma_B}) \overset{d}{=} (D^{\rho}, D^{\sigma_1\rho}, \ldots, D^{\sigma_B\rho}).$$

Because $\mathscr{G}$ is a group and σ_b is uniform and independent of ρ, the elements $\rho, \sigma_1\rho, \ldots, \sigma_B\rho$ are i.i.d. uniform on $\mathscr{G}$. The right-hand side is therefore an exchangeable sequence (conditionally on D, and hence unconditionally). So is the left-hand side.

Step 2: the statistics are exchangeable. Applying the fixed function V coordinatewise preserves exchangeability, so $(V_0, V_1, \ldots, V_B) = (V(D), V(D^{\sigma_1}), \ldots, V(D^{\sigma_B}))$ is exchangeable.

Step 3: the rank of V_0 is super-uniform. For an exchangeable vector, the number $K = \#\{b \geq 0 : V_b \geq V_0\}$ (which counts $b = 0$) satisfies $\Pr(K \leq k) \leq k/(B+1)$ for $k = 1, \ldots, B+1$. Ties only increase K. Since $p_L = K/(B+1)$, $\Pr(p_L \leq \alpha) = \Pr(K \leq \lfloor \alpha(B+1) \rfloor) \leq \lfloor \alpha(B+1) \rfloor/(B+1) \leq \alpha$. □

Remark 1 (What Theorem 1 adds, and what it does not). The argument is the classical one for randomization tests: a test is exact for any statistic fixed before the data are seen, also when the permutations are drawn at random and the identity is counted (Lehmann and Romano 2022; Hemerik and Goeman 2018). A deterministic loop's report is such a statistic, so the theorem adds no new statistical principle. What it adds for AI research loops is the statement of its two hypotheses in checkable form.

1. *Completeness of the replay.* L must be a fixed function of the data. Every choice that depends on outcomes must be inside L, and none outside. Replayability is how this is verified, and it is necessary but not sufficient. In the Monte Carlo, a single pruning step taken after peeking and left out of the replay doubles the size of the test (10.3% instead of 5.0%; Section 7).

2. *Exchangeability* (Assumption 1). The null worlds must have the distribution of the data under the null. Section 8 measures what happens when they do not.

Fixed-family corrections for data snooping (White 2000; Romano and Wolf 2005; Hansen 2005) rest on the same resampling logic, but they need the family of tests enumerated in advance. An adaptive loop's family is itself random. In the 1,000 Type A worlds, the registered loop fits between 2,169 and 5,171 models (median 2,790), and its reported packages use 21 different moderator triples on 53 base specifications. Replay prices the path the loop actually took. Pricing the whole registered garden instead, through the best package anywhere in the design space, is valid but blunter: it gives $p = 0.124$ for the real package, against $p = 0.056$ for the loop's actual stopping rule (`tables/benchmark.json`).

Remark 2 (What adaptivity is allowed, and what breaks validity). The loop may search, stop, prune, rationalize signs, or select outcomes and moderators as adaptively as it likes, because every null world is processed by the *same* function. Validity breaks when some input to the report depends on D through a channel that is not replayed in D^{σ}. Examples are a human or language-model decision made after seeing results ("this moderator looks promising"), or a design space edited after an exploratory run. The registry, frozen and hashed before outcomes were computed, is the device that rules this out. The "leakage" scenario of the Monte Carlo (Section 7) quantifies the distortion when it is violated.

Corollary 1 (Search-adjusted moderator tests)**.** *Fix a moderator k and strata $s(i) = (g(i), q(E_i))$, where $q(\cdot)$ is the treatment quintile. Let $\mathscr{G}_k$ permute the column M_k within strata s. Suppose $D^{\sigma} \overset{d}{=} D$ for $\sigma \in \mathscr{G}_k$, that is, M_k is exchangeable within strata given everything else. Then the statistic $V_k(D)$ (the sign-consistent t of M_k in the reported package, or $-\infty$ if M_k is not reported) yields a valid p-value by Theorem 1.*

Two moderator nulls. The permutation null of Corollary 1, H_{0k}^{perm}, says that M_k carries no information about outcomes within strata. It implies, but is not implied by, the hypothesis the theory states, H_{0k}^{int}: the effect of exposure does not vary with M_k. Permuting M_k also breaks its own outcome trend and its within-stratum correlation with the other moderators (for hierarchy shape, -0.38 with digitizable work inside sector $\times$ exposure-quintile cells). A rejection of H_{0k}^{perm} can therefore come from structure other than the interaction. Firm fixed effects absorb M_k's level, and every moderator model contains $M_k \times$ Post, which absorbs its linear post-shock trend in the real data. Neither term reproduces those relations in the permuted worlds. Null-model worlds (Section 3) target H_{0k}^{int} directly. They keep M_k and all its relations fixed and remove only the interaction. They are not exactly exchangeable: their validity is asymptotic, as for the wild cluster bootstrap (Cameron et al. 2008), and requires the restricted model to contain the structure that the loop's other specifications can detect. Section 8 reports their size and their p-values on the real data.

The statistic. V_k is $-\infty$ in every world in which the loop does not report M_k, so

$$p_k = \Pr(\text{the loop reports } M_k \text{ and } V_k \geq v_k)$$

combines selection and strength. Two further statistics, defined in every world, separate them (Section 8). V_k^{free} is the largest sign-consistent single-moderator t of M_k over the base specifications with a significant H1, the statistic the loop screens on. V_k^{fix} is the t of M_k in the reported specification with its two partner moderators, with no search. Both were added in response to review. They are diagnostics, and no claim rests on them alone.

Familywise error over the ten registered moderators. The loop reports three of ten candidates, so the family is all ten. For an unreported candidate $V_k(D) = -\infty$ and $p_k = 1$. Holm's step-down over the ten p-values (seven of them equal to one) therefore controls the familywise error rate in the strong sense whenever each p_k is valid, under any dependence. It multiplies the three reported p-values by 10, 9, and 8. Holm over the three reported p-values alone carries no such guarantee, because those three were selected for being small: the union bound runs over all ten candidates. In the joint worlds of Section 8, scored against one another, Holm over the three reported moderators confirms at least one in 14.4% of worlds in which none of the ten matters. Holm over ten does so in 4.7%. Joint worlds, in which all ten moderators are permuted independently within strata in the same world, give the Westfall–Young adjustments (Westfall and Young 1993). These use the dependence among the ten statistics. Single-step max-T is dominated by the candidate with the heaviest null tail. Step-down min-P puts the candidates on a common scale; its strong control requires subset pivotality.

Remark 3 (Potemkin rate). For a procedure P that maps data to a binary "fully significant package" verdict, its Potemkin rate is $\Pr_\sigma(P(D^\sigma) = 1)$ over Type A null worlds. It is the procedure's false-discovery probability for the whole package under the conditional null, and it is estimated with Monte Carlo standard error $\sqrt{r(1-r)/B}$.

2 The replayable loop and the estimation engine

Design space (registered). There are three outcome proxies, three exposure proxies, three control sets (none; eight lagged firm controls; the same plus pre-period size and hiring growth interacted with year), two fixed-effect structures (firm and year; firm and NAICS-3 $\times$ year), three sample rules (at least 10 hires; at

least 30 hires; firms observed at least 8 years), two transformations (none; winsorizing at the 1st and 99th percentiles by year), and two clusterings (firm; NAICS-3). That gives 648 base specifications. Inside each, the loop searches the ten registered moderators. The full registry is reproduced in Section 9.

Policies.

- **U** (registered signs) visits base specifications lexicographically and fits H1. If H1 is significant with its registered sign, U fits each moderator singly and keeps those significant with their registered sign. It then fits up to ten mixed-sign triples, ordered by the sum of single $|t|$, and stops at the first package with all four sign-consistent $|t| \geq 1.96$.
- **U-free** is identical except that moderator signs are unrestricted. This models an agent that rationalizes the signs after seeing them.
- **Other procedures.** Bonferroni divides α by the number of models U visited. The specification-curve criterion requires the median H1 t across all base specifications to exceed 1.96. The guard runs U on half of the firms (stratified by NAICS-2, registered seed) and re-estimates the selected package once on the other half, with Holm's correction across the four hypotheses.

Engine. Fixed effects are removed by the method of alternating projections (Gaure 2013), vectorized across columns with sparse indicator matrices. Controls are partialled out once per (sample, fixed effects, transformation, control set) by an orthonormal basis (Frisch–Waugh–Lovell), so each model involves only its one to seven focal columns. Cluster-robust variances use the small-sample factor $\frac{G}{G-1}\frac{N-1}{N-K}$, with K counting the partialled controls.

Validation. Coefficients and standard errors match `pyfixest` to $\leq 4 \times 10^{-15}$ on a simulated unbalanced panel (`intermediate/90_engine_validation.csv`). The FWL path matches direct regressions on the real panel to $\leq 6 \times 10^{-15}$ (`intermediate/91_fwl_validation.csv`). Evaluating both policies in one pass reproduces separate runs exactly.

3 Null worlds and planted worlds

Type A (treatment null). All three exposure proxies are permuted *jointly* across firms within 2-digit NAICS sectors, using world seed $1000003b + 17$.

- **Preserved:** every other variable, the panel structure, missingness, serial dependence, the correlations among exposure proxies, and the sector-level mean of exposure. Sector-level co-movement between exposure and outcomes therefore remains; the null is conservative about *firm-specific* effects.
- **Destroyed:** only the link between a firm's own exposure and its own outcomes.

Type B (moderator null). One moderator is permuted within NAICS-2 $\times$ exposure-quintile cells. The treatment and the other moderators are held fixed.

Constructions added in response to review (`scripts/22_validity_worlds.py`; seeds follow the registered rule with new streams):

- **Covariate-preserving Type A.** Within NAICS-2, each exposure proxy is split into its linear projection on twelve predetermined traits (the ten moderators, pre-period size, and pre-period hiring growth) and a residual. Only the residual is permuted, jointly for the three proxies. Exposure keeps its within-sector correlation with the observed traits. The projection explains 53% of the within-sector variance of the main proxy.

- **Joint Type B.** All ten moderators are permuted independently within NAICS-2 $\times$ exposure-quintile cells in the same world, as the registry's Type B entry describes. This gives the familywise null distribution.
- **Conditional Type B.** M_k is split into its projection on (NAICS-2, the three exposure proxies, exposure^2, the other nine moderators) and a residual, and only the residual is permuted within the Type B cells.
- **Null-model (Freedman–Lane) worlds** for moderator k (Freedman and Lane 1983; Winkler et al. 2014). For each outcome, the restricted model is fitted on the analysis sample: firm and NAICS-3 $\times$ year effects, the eight C1 controls, the three exposure proxies $\times$ Post, exposure$^2\times$Post, all ten $M_j\times$Post, and $E \times M_j\times$Post for $j \neq k$, everything except the target interaction. Each world adds the fitted values to the restricted residuals, with each firm's residuals multiplied by one Rademacher sign shared across outcomes. This keeps within-firm serial dependence and heteroskedasticity.

Planted worlds. Starting from a Type A world, we add $\beta^* \, \mathrm{sd}_o^w \, d_o \, E_i^\sigma \mathrm{Post}_t$ to each outcome o, where sd_o^w is the within-firm standard deviation and d_o is the registered direction. We also add γ^* times the product of that term with each of the three moderators in the real package, in their registered directions. The reported effects are then compared with the planted ones.

4 Data, linkage, and measurement audits

4.1 Sources

- **Revelio Labs positions** are a yearly active-state panel of résumé-based positions (file Y holds positions active in Y). Hires are positions whose start date falls in Y. Seniority is Revelio's seven-level scale.
- **Revelio User Skills** consist of 283 shards (about 850 GB), which we streamed shard by shard.
- **Revelio job postings, 2018–2022**, supply a role-to-occupation crosswalk.
- **Occupational exposure scores** are the public releases of Eloundou et al. (2024) and Felten et al. (2021, 2023).
- **Compustat North America** (quarterly release through 2026Q2) supplies fiscal-year fundamentals through FY2025.
- **I/B/E/S detail** supplies analyst coverage.

Firms are linked through Revelio's company mapping to Compustat identifiers: 11,182 linked firms, each gvkey mapped to one Revelio entity, with listed subsidiaries assigned before their parents. Revelio headcount correlates 0.89 with Compustat employment (logs), at 0.89–0.90 in every year from 2016 to 2023 (`intermediate/07b_headcount_validity.csv`).

4.2 Audits that shaped the measures

- **Skill taxonomy.** Revelio's canonical skill label is unreliable for generative AI. Only 2.6% of rows labelled "GitHub Copilot" carry a GitHub-Copilot string; the rest are plain "github." "Hugging Face

Table A1: Construction of the analysis sample

Step	Firm-years	Firms
Revelio firm-years at Compustat-linked firms (2015-2025)	118,346	11,182
Firm-years 2016-2025	107,676	11,161
with industry code	107,636	11,157
with predetermined exposure (2019-2021 role mix)	107,141	10,810
with >= 10 hires (S1 outcome definition)	67,099	8,425
with non-missing lagged controls (analysis sample S1)	41,352	6,468
observed both before and after the shock	34,333	4,078

Transformers" is 20% valid. Machine translation renders "generative ai" as "generative to." We therefore match an explicit dictionary (51 generative-AI strings, 39 core-AI strings) against the raw string first and the translation second, yielding 3.9 million generative-AI skill rows.

- **Skill dates.** The first-reported dates of generative-AI skills cluster in 2024–2026 (1,779 in 2023; 1.18 million in 2024). They mark résumé capture, not acquisition, so the "ever listed" share (0.7% already in 2016) is a composition measure. We therefore use the dated share as the adoption variable and discuss its timing.

- **Role vocabulary.** Position roles and posting roles share the same 1,500-value vocabulary (1,498 of 1,498 match), allowing a posting-weighted crosswalk that covers 96% of role mass for the Eloundou et al. scores.

Table A2: Theoretical constructs and their measurement

Variable	Type	Definition and formula	Rationale (with citations)
Outcome construct: experience tilt of hiring (firm i, year t)			
Experienced-hire share	Dependent (H1 primary)	$\text{ExpShare}_{it} = \sum_{k\geq 3} H_{itk} / \sum_{k=1}^{7} H_{itk}$, H_{itk} = hires (positions starting in t) at Revelio seniority k (1 entry … 7 senior executive)	Seniority of entry is the organizational margin through which substitution for novices appears (Autor 2024; Brynjolfsson, Chandar & Chen 2025)
Mean hire seniority	Dependent (alt.)	$\sum_k k H_{itk} / \sum_k H_{itk}$	Continuous version of the same construct; less sensitive to the level-3 cut-off
Entry-level hiring rate	Dependent (alt.; registered sign −)	$100 \times (H_{it1} + H_{it2}) / \text{Headcount}_{i,t-1}$	Ports-of-entry view of internal labor markets (Doeringer & Piore 1971)
Experienced-hire share, U.S. hires	Dependent (robustness)	ExpShare computed on U.S.-located positions only	Removes offshore hiring composition
Treatment construct: predetermined workforce exposure to large language models			
LLM exposure (GPT-4 β)	Independent (primary)	$E_i = \sum_r s_{ir} E_r$, s_{ir} = firm's 2019–2021 headcount share in role r; $E_r = \sum_o p(o \mid r)\, \beta_o$, $p(o \mid r)$ = share of role-r postings (2018–2022) in O*NET occupation o; β_o = GPT-4-rated exposure	Task-based exposure of occupations to LLMs (Eloundou et al. 2024); shift-share logic (Borusyak, Hull & Jaravel 2022)
LLM exposure (human β)	Independent (alt.)	as above with human-rated β_o	Convergent validity across raters (Eloundou et al. 2024)
Language-modeling AIOE	Independent (alt.)	as above with Felten et al. language-modeling AIOE	Independent exposure method (Felten, Raj & Seamans 2021, 2023)
Non-language AI exposure	Negative control	residual of general AIOE on language-modeling AIOE across roles, aggregated as E_i	Specificity: exposure to non-language AI should not reproduce the post-2022 break
Post-ChatGPT	Shock	$\text{Post}_t = \mathbb{1}[t \geq 2023]$ (ChatGPT released 30 Nov. 2022)	Timing external to firms

Variable	Type	Definition and formula	Rationale (with citations)
Endogenous adoption (two-stage least squares)			
GenAI adoption (dated)	Endogenous	% of active workforce in t with a generative-AI skill first reported by 31 Dec. t (dictionary of 51 raw skill strings)	Direct measure of workforce uptake; dates track résumé capture (audited)
GenAI listing (ever)	Descriptive	% of active workforce listing any generative-AI skill	Composition, not timing (audited; not used as treatment)
Moderator pool (all predetermined, standardized; registered sign relative to H1)			
Hierarchy shape	Moderator (amplifies)	$\ln\left[(N_1+N_2)/\sum_{k\geq 4} N_k\right]$, headcount by seniority, 2019–2021	Bottom-heavy hierarchies hold more codified junior task mass
Digitizable work	Moderator (amplifies)	mean Revelio remote-suitability of active positions, 2019–2021	Digitally performed tasks are easier to reallocate to AI
Organizational age	Moderator (attenuates)	$\ln(1+2022-\text{founding year})$	Structural inertia (Hannan & Freeman 1984; Sørensen & Stuart 2000)
Internal labor market	Moderator (attenuates)	share of manager-and-above hires 2016–2019 filled by internal movers	Make-versus-buy (Bidwell & Keller 2014)
Financial slack	Moderator (attenuates)	cash/assets, FY2019–2021	Slack buffers pressure for savings (Cyert & March 1963; Bourgeois 1981)
Analyst coverage	Moderator (amplifies)	$\ln(1+\text{analysts})$, FY1 EPS forecasters, 2019–2021	Capital-market short-termism (Graham, Harvey & Rajgopal 2005; He & Tian 2013)
Prior AI talent	Moderator (amplifies)	share of 2019–2021 workforce listing core (non-generative) AI skills	Absorptive capacity (Cohen & Levinthal 1990)
Junior labor cost	Moderator (amplifies)	ln mean salary, seniority 1–2, 2019–2021	Larger savings from substitution
Industry concentration	Moderator (attenuates)	NAICS-3 sales Herfindahl, FY2019–2021	Weaker competitive pressure to adopt
Knowledge intensity	Moderator (attenuates)	R&D/assets, FY2019–2021	Tacit knowledge transmitted by apprenticeship (Kogut & Zander 1992)
Controls (lagged one year) and trend adjustments			
Log headcount	Control	$\ln \text{Headcount}_{i,t-1}$ (Revelio)	Size (Becker & James tests)
Hiring rate	Control	$H_{i,t-1}/\text{Headcount}_{i,t-1}$	Hiring intensity and composition
Separation rate	Control	$\text{separations}_{i,t-1}/\text{Headcount}_{i,t-1}$	Replacement hiring
Log assets	Control	$\ln \text{AT}_{i,t-1}$ (Compustat)	Scale
ROA	Control	OIBDP/AT, $t-1$	Performance pressure
Leverage	Control	(DLTT+DLC)/AT, $t-1$	Financial constraint
Sales growth	Control	$\ln \text{SALE}_{t-1} - \ln \text{SALE}_{t-2}$	Demand shocks
R&D intensity	Control	XRD/AT, $t-1$ (missing = 0)	Innovation intensity
Pre-period size × year	Trend control (C2)	mean ln headcount 2019–2021 × year dummies	Size-specific trends
Pre-period hiring growth × year	Trend control (C2)	$\ln(1+H_{2021}) - \ln(1+H_{2019})$ × year dummies	2021–22 hiring boom
Design variables			
Hires	Sample rule	$\sum_k H_{ik} \geq 10$ (S1), ≥ 30 (S2)	Precision of share outcomes
Industry	Fixed effects / strata	NAICS-3 × year fixed effects; NAICS-2 permutation strata	Absorbs sector shocks (2022–23 technology downturn)

5 The testbed: theory, estimates, identification

5.1 Hypotheses (registered before estimation)

Hypothesis 1 (H1). Firms whose pre-shock workforce was more exposed to large language models shift their hiring toward experienced workers after the release of ChatGPT.

Mechanism. Managers staffing work decide whether a task goes to a new hire, an experienced worker using the model, or the model alone. Generative AI lowers the cost of first drafts, code, summaries, and routine analysis, the codified outputs through which entry-level hires contribute and learn (March 1991;

Table A3: Descriptive statistics (firm-years, 2016–2025)

Variable	N	Mean	SD	P25	Median	P75
Experienced-hire share	34,333	0.401	0.160	0.289	0.383	0.500
Mean hire seniority	34,333	2.573	0.552	2.207	2.519	2.896
Entry-level hires per 100 employees	34,333	9.005	6.482	4.951	7.830	11.321
LLM exposure (GPT-4 β)	34,333	0.473	0.076	0.423	0.473	0.523
GenAI adoption (% of workforce)	34,333	0.084	0.376	0.000	0.000	0.000
Internal labor market	34,333	0.373	0.135	0.281	0.375	0.465
Financial slack	34,333	0.184	0.200	0.048	0.108	0.242
Analyst coverage (log)	34,333	1.619	1.105	0.693	1.792	2.512
Organizational age (log)	34,333	3.642	0.807	3.135	3.611	4.220
Digitizable work	34,333	0.499	0.123	0.434	0.520	0.583
Prior AI talent	34,333	0.017	0.023	0.004	0.008	0.019
Hierarchy shape (log ratio)	34,333	0.825	0.644	0.432	0.866	1.225
Junior labor cost (log)	34,333	10.875	0.322	10.738	10.908	11.066
Industry concentration	34,333	0.089	0.102	0.025	0.053	0.105
Knowledge intensity	34,333	0.037	0.093	0.000	0.000	0.025
Log headcount$_{t-1}$	34,333	7.851	1.889	6.423	7.814	9.202
Hiring rate$_{t-1}$	34,333	0.150	0.079	0.096	0.135	0.186
Separation rate$_{t-1}$	34,333	0.119	0.055	0.081	0.111	0.147
Log assets$_{t-1}$	34,333	7.502	2.241	6.102	7.530	8.963
ROA$_{t-1}$	34,333	0.034	1.113	0.020	0.080	0.136
Leverage$_{t-1}$	34,333	0.299	1.359	0.082	0.243	0.411
Sales growth$_{t-1}$	34,333	0.097	0.428	-0.023	0.069	0.184
R&D/assets$_{t-1}$	34,333	0.041	0.119	0.000	0.000	0.027

Notes. Firm-level variables (exposure and moderators) are repeated across a firm's years. Moderators are shown before standardization.

Levinthal and March 1993). It does not replace the judgment needed to evaluate and integrate those outputs. The marginal hire therefore moves up the ladder.

Rival prediction. Experiments find the largest productivity gains for novices (Noy and Zhang 2023; Brynjolfsson et al. 2025; Dell'Acqua et al. 2026). That implies firms could substitute AI-assisted novices for expensive experienced hires.

The moderators the loop reported are three corollaries of the H1 mechanism, registered with signs before estimation:

- **Hypothesis 2 (H2): hierarchy shape amplifies.** The more bottom-heavy a firm's hierarchy before the shock, the stronger the shift away from entry-level hiring. A pyramid holds more of the codified junior task mass that AI can absorb, because a knowledge hierarchy solves its routine problems at the bottom (Garicano 2000).

- **Hypothesis 3 (H3): digitizable work amplifies.** The more digitizable a firm's work, the stronger the shift. Tasks already performed digitally are easier to reassign to AI tools, since technology substitutes for codifiable tasks (Autor et al. 2003).

- **Hypothesis 4 (H4): organizational age attenuates.** The older the organization, the weaker the shift. Institutionalized staffing routines and structural inertia slow redesign (Hannan and Freeman 1984; Sørensen and Stuart 2000).

The registry states the H2 and H3 mechanisms without a citation. The citations above name the theories those mechanisms apply (knowledge hierarchies; task-based technological change) and were added in re-

vision on 27 October 2026; no hypothesis, direction, measure, or test changed. The other seven registered moderators and their estimates are reported in Table A7.

5.2 Estimates

Table A4: Generative-AI exposure and the experience tilt of hiring (H1)

	(1) Exp. share	(2) Exp. share	(3) Exp. share	(4) Mean sen.	(5) Entry rate	(6) Entry rate
Exposure × Post (H1)	1.3172***	1.2066***	0.7353***	0.0162***	-0.2825***	-0.2005**
	(0.1220)	(0.1250)	(0.2276)	(0.0041)	(0.0861)	(0.0848)
	[p<0.001]	[p<0.001]	[p=0.00124]	[p<0.001]	[p=0.00105]	[p=0.0181]
Firm controls (lagged)	No	Yes	Yes	Yes	Yes	Yes
Pre-period size, hiring growth × year	No	No	Yes	No	No	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	–	Yes	–	–
Industry (NAICS-3) × year FE	No	No	Yes	No	Yes	Yes
Clustered SE	Firm	Firm	Firm	Firm	Firm	Firm
Observations	34,333	34,333	34,333	34,333	34,333	34,333
Firms	4,078	4,078	4,078	4,078	4,078	4,078
Within R^2	0.008	0.016	0.012	0.015	0.158	0.256
Pre-period mean of outcome	36.698	36.698	36.698	2.495	10.516	10.516
1-SD exposure effect, % of pre-period mean	+3.6%	+3.3%	+2.0%	+0.6%	-2.7%	-1.9%

Notes. Unit: firm-year, 2016–2025. Exposure is the firm's 2019–2021 role-mix-weighted GPT-4 β exposure (Eloundou et al. 2024), standardized across firms; Post = 1 for 2023–2025. Outcomes: share of hires at associate level or above (%, columns 1–3); mean seniority of hires (1–7, column 4); entry-level hires per 100 employees (winsorized at 1/99 by year, columns 5–6; the registered H1 sign is negative). Standard errors clustered by firm in parentheses; two-sided p-values in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Economic magnitudes.

- **H1.** A one-standard-deviation higher exposure raises the experienced-hire share by 1.21 percentage points (3.3% of its pre-period mean).
- **Package.** In the package the loop reported, the same difference lowers entry-level hiring by -1.08 hires per 100 employees at average moderator values.
- **Moderators.** The reduction is 13.5% of the pre-period mean for firms one standard deviation more bottom-heavy than average, against 6.9% one standard deviation less. For digitizable work the contrast is 13.2% against 7.3%. For organizational age it is 9.0% for older firms against 11.5% for younger ones.
- **IV.** Under the exclusion restriction, one percentage point of workforce generative-AI adoption raises the experienced-hire share by 6.5 points (SE 0.7).

5.3 Identification

A measurement caveat. Raw means (Figure A1) show the experienced-hire share rising after 2022 in all exposure terciles. Part of this common rise likely reflects the lag with which recent hires, disproportionately junior, add positions to online profiles. Year fixed effects absorb the common component. A bias in the exposure coefficient would require the reporting lag to differ systematically by exposure. The estimate is

Table A5: Organizational moderators of the exposure effect on entry-level hiring (H2–H4)

	(1) H1	(2) +H4	(3) +H3	(4) +H2	(5) Joint	(6) Joint+C2
Exposure × Post (H1)	-0.2825***	-0.2349***	-0.7247***	-1.0005***	-1.0758***	-1.0958***
	(0.0861)	(0.0848)	(0.1006)	(0.0953)	(0.1014)	(0.1012)
	[p=0.00105]	[p=0.00564]	[p<0.001]	[p<0.001]	[p<0.001]	[p<0.001]
Exposure × Post × Organizational age (H4)		0.1070**			0.1317***	0.0997**
		(0.0532)			(0.0507)	(0.0490)
		[p=0.0444]			[p=0.0094]	[p=0.0421]
Exposure × Post × Digitizable work (H3)			-0.1198**		-0.3090***	-0.2827***
			(0.0584)		(0.0675)	(0.0646)
			[p=0.0401]		[p<0.001]	[p<0.001]
Exposure × Post × Hierarchy shape (H2)				-0.1178**	-0.3462***	-0.3131***
				(0.0509)	(0.0621)	(0.0614)
				[p=0.0208]	[p<0.001]	[p<0.001]
Organizational age × Post		0.4522***			0.6078***	0.5584***
		(0.0501)			(0.0498)	(0.0517)
		[p<0.001]			[p<0.001]	[p<0.001]
Digitizable work × Post			0.7213***		0.3038***	0.3680***
			(0.0994)		(0.0976)	(0.0947)
			[p<0.001]		[p=0.00187]	[p<0.001]
Hierarchy shape × Post				-1.1817***	-1.1620***	-1.2240***
				(0.0752)	(0.0769)	(0.0774)
				[p<0.001]	[p<0.001]	[p<0.001]
Firm controls (lagged)	Yes	Yes	Yes	Yes	Yes	Yes
Pre-period size, hiring growth × year	No	No	No	No	No	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry (NAICS-3) × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Clustered SE	Firm	Firm	Firm	Firm	Firm	Firm
Observations	34,333	34,333	34,333	34,333	34,333	34,333
Firms	4,078	4,078	4,078	4,078	4,078	4,078
Within R^2	0.158	0.161	0.161	0.172	0.180	0.279
Pre-period mean of outcome	10.516	10.516	10.516	10.516	10.516	10.516
1-SD exposure effect, % of pre-period mean	-2.7%	-2.2%	-6.9%	-9.5%	-10.2%	-10.4%

Notes. Outcome: entry-level hires per 100 employees (winsorized at 1/99 by year); a negative coefficient on Exposure × Post means fewer entry-level hires. Moderators are measured before the shock (2016–2021) and standardized across firms; their main effects are absorbed by firm fixed effects, and Moderator × Post terms absorb differential trends. Registered signs: hierarchy shape amplifies (more negative), digitizable work amplifies, organizational age attenuates (less negative). Column 5 is the package reported by the registered loop (policy U) after 3,483 models; search-adjusted p-values are in Table A11. Standard errors clustered by firm. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A6: Workforce generative-AI adoption: first stage and two-stage least squares

	(1) GenAI adopt.	(2) Exp. share	(3) Exp. share	(4) Entry rate	(5) Entry rate
Exposure × Post (H1)	0.1846***	1.2066***		-0.0465	
	(0.0078)	(0.1250)		(0.0518)	
	[p<0.001]	[p<0.001]		[p=0.37]	
GenAI adoption (%)			6.5358***		-0.2519
			(0.7348)		(0.2799)
			[p<0.001]		[p=0.368]
Firm controls (lagged)	Yes	Yes	Yes	Yes	Yes
Pre-period size, hiring growth × year	No	No	No	No	No
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	2SLS	OLS	2SLS
Clustered SE	Firm	Firm	Firm	Firm	Firm
Observations	34,333	34,333	34,333	34,333	34,333
Firms	4,078	4,078	4,078	4,078	4,078
Within R^2	0.127	0.016	–	0.169	–
Pre-period mean of outcome	0.000	36.698	36.698	10.516	10.516
1-SD exposure effect, % of pre-period mean	–	+3.3%	–	-0.4%	–

Notes. Column 1: first stage; the outcome is the percentage of the year-t workforce with a dated generative-AI skill (first-stage F = 562). Columns 2 and 4: reduced forms; columns 3 and 5: two-stage least squares instrumenting adoption with Exposure × Post. The exclusion restriction requires exposure to affect hiring only through workforce adoption; the reduced forms remain intent-to-treat estimates if it fails. Firm and year fixed effects; standard errors clustered by firm.

Table A7: The registered moderator pool: single-moderator estimates on the primary outcome

Moderator	Registered sign	Coef. (pp)	(SE)	Sign-consistent t
Internal labor market	attenuates	0.200	(0.140)	-1.42
Financial slack	attenuates	0.034	(0.193)	-0.18
Analyst coverage (log)	amplifies	-0.302	(0.123)	-2.45
Organizational age (log)	attenuates	-0.246	(0.126)	+1.96
Digitizable work	amplifies	-0.174	(0.094)	-1.86
Prior AI talent	amplifies	-0.399	(0.181)	-2.21
Hierarchy shape (log ratio)	amplifies	0.543	(0.106)	+5.15
Junior labor cost (log)	amplifies	-0.028	(0.108)	-0.26
Industry concentration	attenuates	0.088	(0.119)	-0.74
Knowledge intensity	attenuates	-0.604	(0.284)	+2.12

Notes. Outcome: experienced-hire share (pp). Each row adds one moderator's interaction with exposure × post and its moderator × post trend to the H1 model (firm and year fixed effects, lagged controls, SE clustered by firm). A positive sign-consistent t means the estimate has the registered direction; negative values are significant *against* the registered theory, which an unconstrained loop could rationalize ex post.

robust to excluding 2025, the year most affected by reporting lags (Table A8), and the break appears already in 2023.

Table A8: Identification checks for H1

Check	Coef. (pp)	(SE)	t	N
Placebo shock in 2019 (pre-period data only)	-0.26	(0.10)	-2.73	23,403
Placebo shock in 2020 (pre-period data only)	-0.24	(0.10)	-2.39	23,403
Negative control: non-language AI exposure × Post	-0.44	(0.12)	-3.62	34,333
Negative control, jointly with LLM exposure	-0.63	(0.12)	-5.36	34,333
LLM exposure, jointly with negative control	1.30	(0.13)	10.28	34,333
Excluding the information sector (NAICS 51)	1.17	(0.14)	8.22	30,420
Remote suitability × year and NAICS-3 × year FE	1.22	(0.29)	4.26	34,333
Excluding 2025 (new data delivery)	0.99	(0.13)	7.95	30,865
Excluding 2020–2021 (COVID-19)	1.12	(0.13)	8.57	27,248
U.S.-headquartered firms	1.10	(0.15)	7.42	26,433
Outcome from U.S. hires only	0.82	(0.15)	5.55	30,437
Standard errors clustered by NAICS-3	1.21	(0.27)	4.51	34,333
Firm-years with $\geq$30 hires	1.11	(0.12)	9.14	30,595
Ex post: exposure-specific linear trend	1.38	(0.14)	10.19	34,333
Ex post: 2019–2025 window	1.26	(0.12)	10.24	25,297
Ex post: trend and NAICS-3 × year FE	1.45	(0.24)	5.94	34,333

Notes. Outcome: experienced-hire share in percentage points; coefficient on the (placebo or control) exposure × post term unless stated. All models include firm and year fixed effects and the lagged controls; standard errors clustered by firm. Joint test of no pre-trends (2016–2021 event-study coefficients): $\chi^2(6) = 16.6$, $p = 0.011$; approximate relative-magnitudes breakdown value $\bar{M}^* = 1.51$ (Rambachan and Roth 2023). Rows marked *ex post* were added after the event study was inspected.

Table A9: Event-study coefficients (experienced-hire share, pp)

Year	Coef.	(SE)
2016	0.19	(0.16)
2017	0.34	(0.15)
2018	0.35	(0.14)
2019	0.08	(0.14)
2020	-0.24	(0.15)
2021	-0.00	(0.11)
2022	0.00	(0.00)
2023	1.25	(0.13)
2024	0.81	(0.16)
2025	1.80	(0.18)

Notes. Coefficients on exposure × year indicators; 2022 is the reference year. Firm and year fixed effects, lagged controls, SE clustered by firm.

Table A10: Pre-period characteristics by terciles of LLM exposure

Variable (2019–2021 mean)	Low	Middle	High	Norm. diff. (High−Low)
Experienced-hire share	0.292	0.408	0.440	+1.19
Mean hire seniority	2.204	2.660	2.771	+1.33
Entry-level hires per 100 employees	10.127	10.964	11.252	+0.17
Log headcount$_{t-1}$	7.904	7.430	7.823	-0.04
Hiring rate$_{t-1}$	0.138	0.172	0.183	+0.64
Separation rate$_{t-1}$	0.115	0.129	0.138	+0.49
Log assets$_{t-1}$	7.406	7.038	7.469	+0.03
ROA$_{t-1}$	0.083	-0.030	0.017	-0.33
Leverage$_{t-1}$	0.349	0.304	0.243	-0.25
Sales growth$_{t-1}$	0.024	0.113	0.114	+0.35
R&D/assets$_{t-1}$	0.004	0.072	0.056	+0.69
Internal labor market	0.378	0.361	0.361	-0.13
Financial slack	0.106	0.245	0.231	+0.77
Analyst coverage (log)	1.625	1.503	1.641	+0.01
Organizational age (log)	3.721	3.594	3.494	-0.28
Digitizable work	0.420	0.536	0.553	+1.15
Prior AI talent	0.006	0.013	0.032	+1.14
Hierarchy shape (log ratio)	1.293	0.633	0.459	-1.59
Junior labor cost (log)	10.704	10.960	10.987	+0.91
Industry concentration	0.114	0.070	0.084	-0.29
Knowledge intensity	0.004	0.069	0.053	+0.72

Notes. Normalized difference = difference in means divided by the square root of the average of the two variances. Level differences are absorbed by firm fixed effects; identification rests on parallel trends (Table A8, Figure A1).

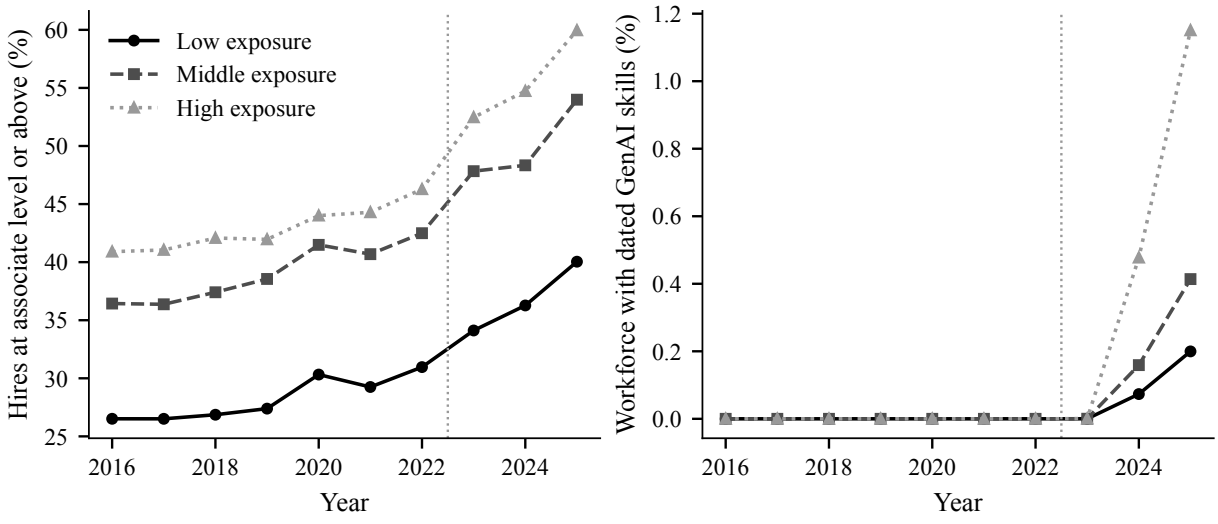


Figure A1: Raw means by exposure tercile: experienced-hire share (left) and workforce generative-AI adoption (right). The dotted line marks the release of ChatGPT.

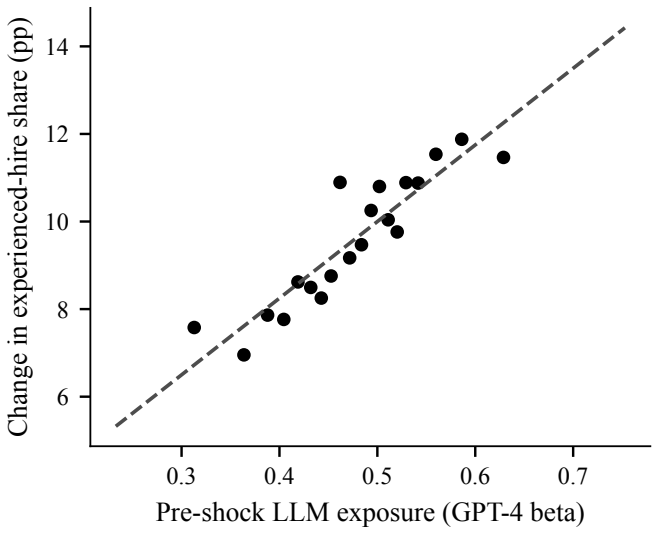


Figure A2: Bin scatter (20 bins) of each firm's post-minus-pre change in the experienced-hire share against its pre-shock LLM exposure, with the OLS line.

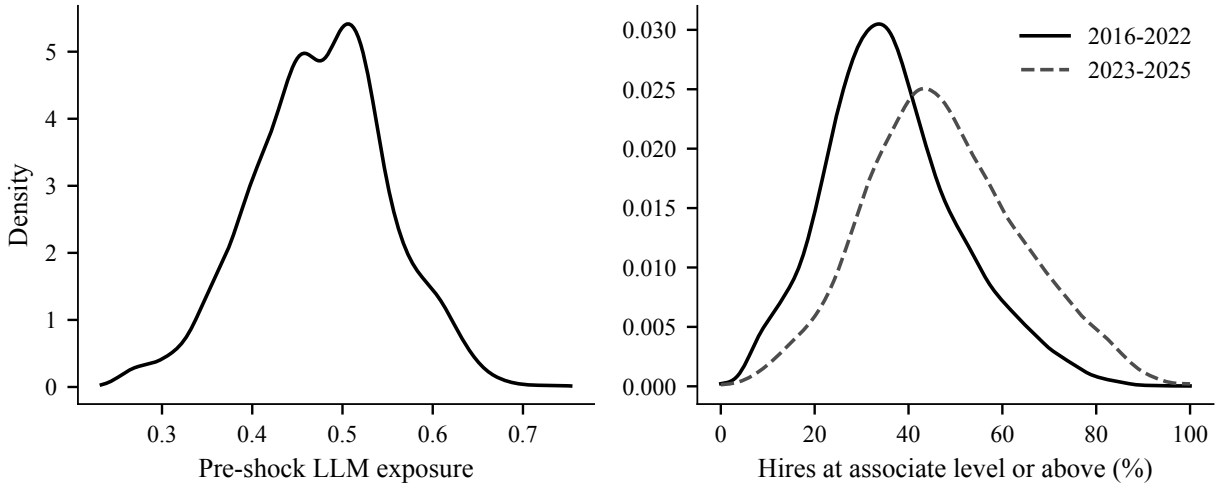


Figure A3: Kernel densities of pre-shock exposure (left) and of the experienced-hire share before and after the shock (right).

6 Calibration results

Reading the results.

- **Null worlds are not quiet.** Across 1,000 Type A worlds, H1 is nominally significant in 34% of the base specifications on average, although exposure has no firm-specific effect by construction. Sector-level co-movement, which within-sector permutation preserves, is enough to generate large t-statistics under firm and year fixed effects. The 95th percentile of the loop's maximum H1 t in null worlds is 8.6.

Table A11: Potemkin rates and search-adjusted inference

Procedure	Rate / p	Interval / adjusted
U (registered signs)	26.5	[23.8, 29.2]
U-free (signs rationalized)	93.4	[91.9, 94.9]
Bonferroni over models visited	0.0	[0.0, 0.0]
Specification curve (median t >= 1.96)	3.3	[2.2, 4.4]
NWC (loop-level p <= .05)	5.0	[3.6, 6.4]
Guard: split sample, one confirmation, Holm	0.6	[0.0, 1.3]
Search-adjusted p-values for the real report		
H1 (maximum t over the search)	0.001	
Package reported by loop U	0.056	
Moderator Organizational age (log) (Type B, B=500)	0.313	Holm over ten: 1.000
Moderator Digitizable work (Type B, B=500)	0.046	Holm over ten: 0.413
Moderator Hierarchy shape (log ratio) (Type B, B=500)	0.002	Holm over ten: 0.020

Notes. Top panel: share (%) of 1,000 Type A null worlds (LLM exposure permuted across firms within 2-digit NAICS sectors) in which the procedure reports H1 and three moderators all significant at the 5% level with mixed signs; 95% Monte Carlo intervals. Bottom panel: rank of the real statistic among the null-world statistics produced by the same loop (Theorem 1). Holm over ten: step-down adjustment over all ten registered moderators, the seven the loop did not report counted as $p = 1$ (Table A17).

- **Potemkin rates.** The unguarded loop reports a full package in 26.5% of null worlds. Letting it rationalize moderator signs raises this to 93.4%.
- **Search-adjusted inference.** The real data's maximum H1 t (12.1) exceeds every null world, and the reported package has search-adjusted $p = 0.056$.

An ex post diagnostic for the moderators. In Type B worlds for digitizable work, the loop still selects the *permuted* moderator in most worlds. Permuting within sector $\times$ exposure-quintile strata preserves the moderator's stratum-level association with exposure, so part of what the loop reads as moderation by digitizable work is heterogeneity of the exposure effect across sectors and exposure levels. Consistent with this, adding exposure$^2 \times$Post to the reported package shrinks the digitizable-work interaction from $t = -4.40$ to $t = -2.95$, while the hierarchy-shape ($t = -5.21$) and organizational-age ($t = 2.30$) interactions are unchanged (`tables/expost_nonlinearity.json`). This diagnostic was run after the Type B results were seen and is labelled accordingly.

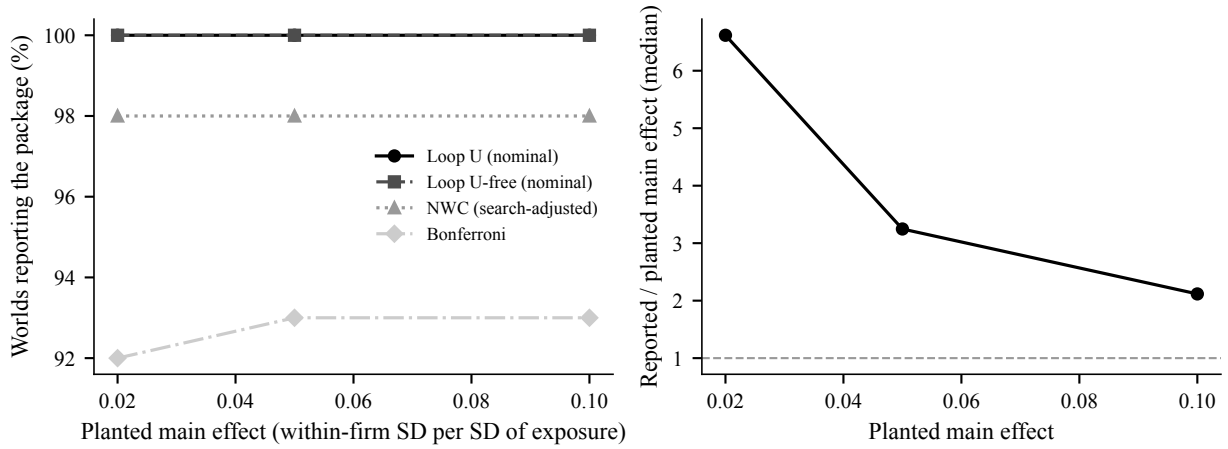


Figure A4: Planted worlds: share of worlds in which each procedure reports the full package (left), and the median ratio of the main effect reported by loop U to the planted effect (right). Planted moderators: 0.05 within-firm SD in their registered directions.

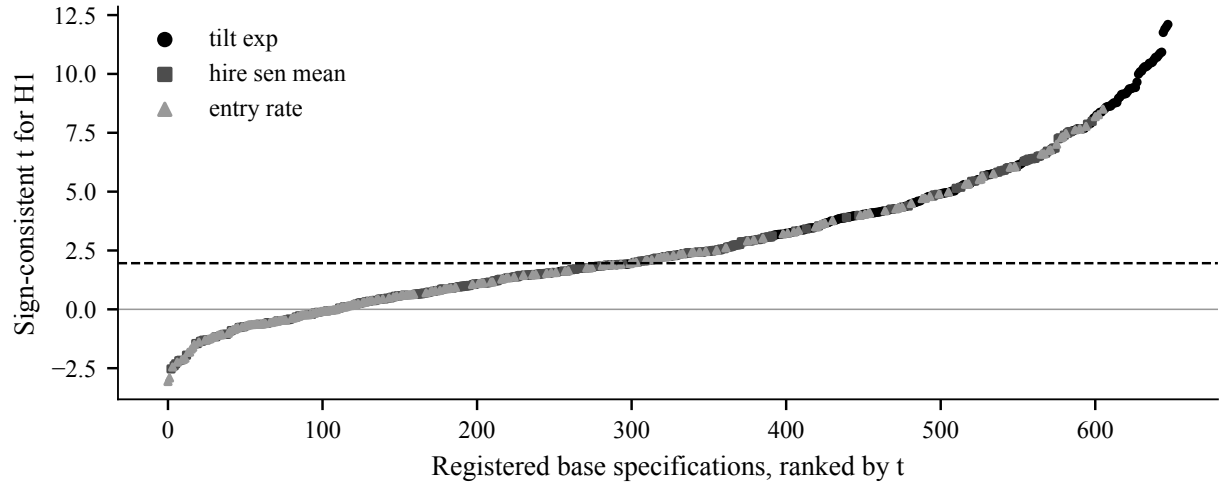


Figure A5: Specification curve: sign-consistent H1 t-statistics across the 648 registered base specifications, by outcome proxy.

7 Monte Carlo evidence on synthetic twins

What the Monte Carlo shows.

- **Size.** Under the null, the loop-level test of the searched main effect rejects in 5.0% of replications, consistent with Theorem 1 given Monte Carlo error. The specification-curve rule rejects in 11.0%: sector-level shocks correlated with exposure make many specifications nominally significant.
- **Proposal leakage.** Suppose the analyst peeks at the results, prunes the design space to the best-looking specification, and only then registers it. The test then rejects in 10.3% of null replications. Replaying the pruning inside every null world restores 5.0%. This is the replayability principle in numbers.

Table A12: Monte Carlo on synthetic twins: rejection rates (%)

Scenario	R	U	Bonferroni	Spec. curve	NWC package	NWC H1
Null (no effect; treatment exchangeable within sector)	200	0.0	0.0	11.0	0.0	5.0
Null, moderators chosen by peeking at outcomes (not replayed)	300	0.0	0.0	10.0	0.0	10.3
Null, the same peeking step replayed in every null world	200	0.0	0.0	11.0	0.0	5.0
No causal effect; unobserved trait drives both exposure and post-period drift	50	0.0	0.0	94.0	4.0	90.0
Planted effects (H1 0.10 SD; three moderators $\pm$0.06 SD)	50	82.0	60.0	100.0	82.0	100.0

Notes. Each replication draws a synthetic twin (Section 7), runs the reduced registered loop (12 base specifications and the 10-moderator search), and computes loop-level p-values from B inner null worlds (within-sector permutation of exposure). Under the null, NWC rejection rates should not exceed 5% up to Monte Carlo error; Theorem 1 does not cover the leakage and confounded scenarios, which show what the method cannot do.

- **Confounding.** When an unobserved trait drives both exposure and post-period outcome drift, the test rejects in 90.0% of replications. Null-World Calibration prices search, not confounding. Section 8 maps how the rejection rate grows with the strength of such a trait, and what restores size when the trait is observed.
- **Power.** With planted effects, the package test rejects in 82.0% of replications and the main-effect test in 100.0%.
- **Twin versus real null worlds.** In the twin, the unguarded loop rarely assembles a full package under the null (0.0%), whereas it does so in 26.5% of real-data null worlds. Real organizational data carry moderator-linked trends and heterogeneous shocks that a simple generative model omits. This is a reason to calibrate on null worlds built from the real data rather than on simulations alone.

The synthetic twin is a generative model fitted to the analysis panel (`nullworld/synth.py`):

- a Gaussian copula over firm-level exposure, moderators, and pre-period covariates;
- sectors drawn conditional on exposure quintile;
- coverage patterns borrowed from real firms;
- outcomes and controls built from firm effects, year effects, sector-year shocks, and AR(1) within-firm shocks.

It reproduces means, between- and within-firm standard deviations, and the firm-level correlations among exposure and moderators. It understates within-firm serial correlation (`intermediate/08_twin_fidelity.csv`).

8 Validity diagnostics

This section examines the conditions under which the calibration is valid: who took which decision, what happens when exposure is not exchangeable within sector, what the moderator tests and their p-values measure (including the familywise correction over all ten registered candidates), and what replay costs. The analyses were added after the main results were seen (20 October 2026). They are diagnostics: they test whether

the calibration's conditions hold and what its p-values measure, and no claim is upgraded on their strength alone. Scripts: `scripts/22–27`; all numbers in `tables/validity.json` and `tables/benchmark.json`.

8.1 Who decided what

Table A13: Who took which research decision in the prototype, and how calibration treats it

Decision	Proposed / taken by	When	Treatment in calibration
Research question, testbed, fit with the call	Authors	Before data access	Fixed; outside the loop
Constructs and proxies (3 outcomes, 3 exposures), controls, fixed effects, samples, transforms, clustering	AI agent, from the literature; approved by the authors	Before any outcome (registry hashed 29 Sep 2026, 04:19)	Frozen and hashed: a non-replayable input that cannot depend on outcomes
Ten moderators with signs and mechanisms; H1 mechanism and rival	AI agent, from the literature; approved by the authors	Before any outcome	Frozen and hashed
Search policies U and U-free, package statistic, stopping rule	Specified in the registry; coded by the AI agent	Before any outcome	Code: replayed identically in every world
Data assembly and measurement audits (skill dictionary, role crosswalk, linkage)	AI agent; reviewed by the authors	Before any outcome	Fixed code; audits reported
Run-time choices: which specification next, which moderators to keep, which triples, when to stop, which signs (U-free)	The coded policy	After data access	Replayed in every null, moderator, and planted world
Ex post identification checks (trend break, 2019+ window, exposure^2 diagnostic)	Authors with the AI agent	After outcomes were seen	Not priced; labelled ex post
Stage 2: run-time choices taken by a live LLM agent	LLM agent (open weights, deterministic decoding)	After data access	Each call logged and re-issued in every world

Notes. Reconstructed from `registry/design_space.yaml` (field `authoring`), the AI-use statement in `README.md`, and the dated entries of `intermediate/00_design_decisions.md`. The AI agent works under the authors' direction.

The AI agent's decisions in the prototype were taken before outcomes were computed, and they are frozen in the registry or the code. All run-time choices are code and are replayed. Policies U and U-free are benchmarks that bracket an agent's search; they are not models of agent behaviour. Figure A6 shows how their Potemkin rates grow with the number of models a loop may fit before reporting.

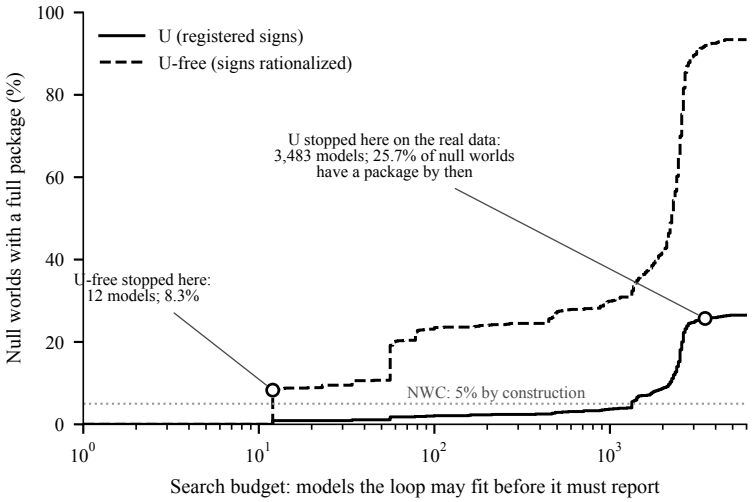


Figure A6: Share of the 1,000 Type A null worlds in which each policy has reported a fully significant package within a given number of models (search budget). Markers: where each policy stopped on the real data.

8.2 When exposure is not exchangeable within sector

Within NAICS-2 sectors, exposure is correlated with several firm traits: 0.62 with digitizable work, -0.53 with hierarchy shape, 0.49 with pre-period AI talent, and 0.16 with pre-period hiring growth. Type A worlds break these links. Table A14 and Figure A7 measure the consequence on synthetic twins with no exposure effect. A trait u with within-sector correlation ρ with exposure shifts outcomes either in levels or after the shock.

- **Level-only traits are harmless.** Firm effects absorb them (6.0% rejections; 6.5% with no trait).
- **Post-shock traits are confounders.** Rejections rise quickly with the spurious effect they induce: 19.5% when it is 5% of the real H1 estimate, 43.0% at 9%, and 100.0% at 26%. A correctly sized test of an association detects a confounded association as readily as a real one: calibration prices search, not confounding. Nominal tests give no protection either. One pre-specified regression already rejects in 17.0% of null twins, because sector shocks correlated with exposure fool its reference distribution, and it rises with confounding too.
- **Observed traits have a remedy.** Strata on the trait's quintile help only partly (up to 45.0%). Worlds that preserve exposure's projection on the trait hold size in all 12 settings (at most 7.5%).

On the real data (Table A15), worlds that preserve exposure's within-sector projection on twelve predetermined traits give $p \leq 0.001$ for H1 and $p = 0.137$ for the package. The unguarded loop's Potemkin rate rises to 40.3%: exposure-correlated traits give a searching loop more ways to assemble a package.

Table A14: Monte Carlo: rejection rates (%) when exposure is not exchangeable within sector

			Spurious	NWC worlds			Nominal	
Trait channel	ρ	δ	effect (%)	within sector	strata on u	u-preserving	one regression	naive loop
level only	0.6	1	1	6.0	6.5	6.0	17.0	20.5
none (null)	0.0	0	1	6.5	5.5	6.0	17.0	19.5
post-shock shift	0.2	0.025	3	11.5	6.0	6.5	20.0	23.5
post-shock shift	0.2	0.05	5	19.5	6.0	7.0	25.0	30.0
post-shock shift	0.2	0.1	9	39.5	9.0	6.0	35.0	42.0
post-shock shift	0.2	0.2	18	87.0	12.0	5.5	58.5	61.5
post-shock shift	0.4	0.025	5	20.0	7.0	6.0	25.5	30.5
post-shock shift	0.4	0.05	9	43.0	8.0	7.0	35.0	43.0
post-shock shift	0.4	0.1	18	87.5	14.5	6.0	61.5	63.0
post-shock shift	0.4	0.2	34	100.0	22.0	5.5	93.5	96.0
post-shock shift	0.6	0.025	7	29.0	9.5	5.5	30.5	35.5
post-shock shift	0.6	0.05	14	70.0	12.5	7.5	48.0	52.5
post-shock shift	0.6	0.1	26	100.0	19.5	5.5	84.5	87.0
post-shock shift	0.6	0.2	51	100.0	45.0	4.5	100.0	100.0

Notes. $R = 200$ synthetic twins per row, $B = 99$ null worlds per twin; Monte Carlo standard error about 1.5 points at 5%. Exposure has no effect. A firm trait u with within-sector correlation ρ with exposure shifts the outcome either in levels (1 between-firm SD) or after the shock (δ within-firm SDs). Spurious effect: the mean estimate of the pre-specified regression as a share of the real H1 estimate. NWC: within-sector permutation (the application's Type A worlds); strata on u: within sector × quintile of u; u-preserving: exposure = projection on (sector, u) + permuted residual.

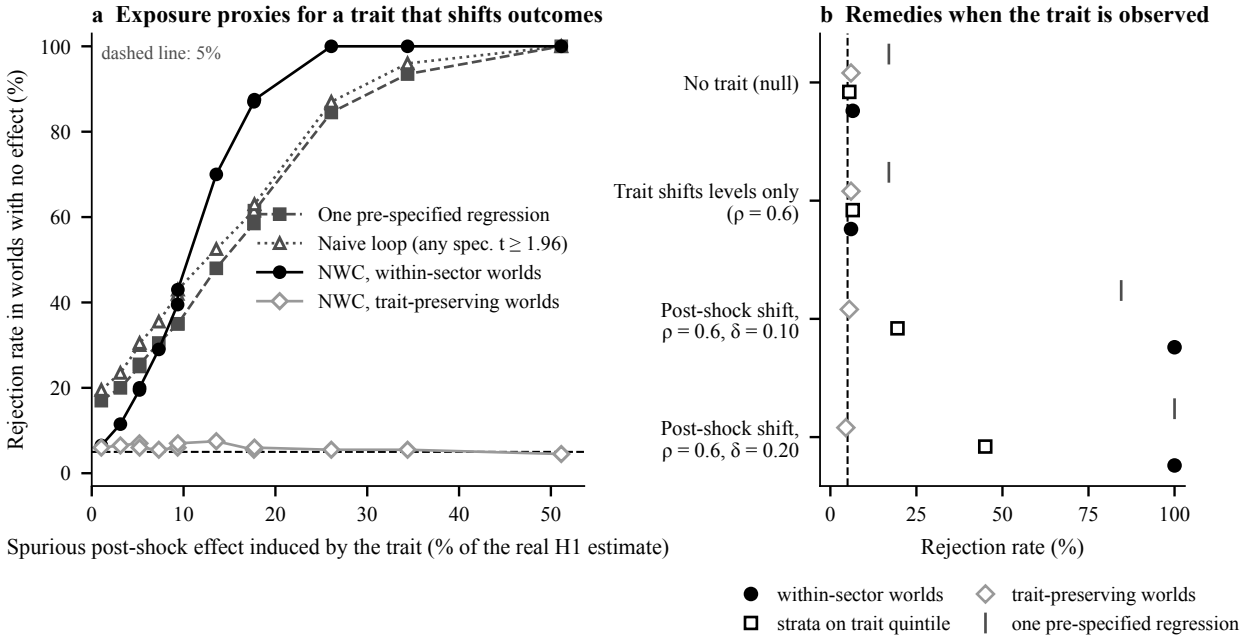


Figure A7: Synthetic twins with no exposure effect and a trait correlated with exposure within sector. (a) Rejection rates against the spurious effect the trait induces (as a share of the real H1 estimate). (b) The null, a level-only trait, and two post-shock cells under four procedures.

8.3 Moderator nulls and what the moderator p measures

Table A16 and Figure A8 report the size of three moderator-world constructions when the target's true interaction is zero. The target is given its own post-shock trend, a correlated moderator with a true interaction,

Table A15: Real data: Type A worlds versus covariate-preserving worlds

	Type A (within sector)	Covariate-preserving
Worlds	1,000	1,000
Largest null H1 t: median / 95th pct. / max	7.4 / 8.6 / 9.8	7.2 / 8.0 / 9.3
Search-adjusted p, H1 (real $t = 12.1$)	0.001	0.001
Search-adjusted p, full package	0.056	0.137
Potemkin rate, U (%)	26.5	40.3
Potemkin rate, U-free (%)	93.4	100.0

Notes. Covariate-preserving worlds keep each exposure proxy's within-sector projection on twelve predetermined traits (the ten moderators, pre-period size and hiring growth; within-sector R^2 53% for the main proxy) and permute only the residual within sector, so trait-driven outcome trends are present in every null world.

and a concave exposure effect, all with adversarial signs. The moderator's own trend is harmless (2.0% rejections under permutation), because every moderator model contains $M_k \times$Post. A correlated moderator with a real interaction and a concave exposure effect are not: permutation rejects a zero interaction in 67.5% and 35.0% of samples, and in 95.0% with all three structures. Conditional permutation holds size in every scenario (4.5% with all three). Null-model worlds are valid but conservative (0.5%); their p-values cluster near 0.5. Table A17 and Figure A9 give the real-data p-values by construction and statistic, with the family-wise corrections over the ten registered candidates. With all ten moderators permuted, the loop still reports a fully significant package in 94.0% of the 1,000 joint worlds. Much of what a loop reads as moderation in these data is the heterogeneity of a strong main effect.

Table A16: Monte Carlo: rejection rates (%) of moderator tests when the true interaction is zero

	With search (S^{free})				One specification (S^{fix})			
Scenario	Nominal	Perm.	Cond.	Null m.	Nominal	Perm.	Cond.	Null m.
No other structure	4.5	4.0	7.0	0.5	2.5	4.0	5.0	0.5
Own post-shock trend of M_k	3.5	1.5	5.5	1.0	2.5	2.0	5.0	0.5
Correlated moderator, true interaction	96.0	65.5	5.0	0.5	95.0	67.5	5.0	0.5
Concave exposure effect	96.5	35.5	5.5	0.5	96.0	35.0	6.0	0.5
All three	96.5	91.5	4.5	0.5	100.0	95.0	4.5	0.5

Notes. $R = 200$ synthetic twins per row with a planted main effect (0.10 within-firm SD), $B = 59$ worlds per construction. Target: hierarchy shape, whose true interaction is zero. Own trend: $+0.10$ SD $\times M_k \times$Post; correlated moderator: digitizable work ($r = -0.60$ with M_k) with interaction -0.06 SD; concave exposure: -0.05 SD $\times (E^2 - 1) \times$Post. Signs are adversarial: the leakage runs in M_k's registered direction. Perm.: within sector $\times$ exposure quintile (Type B); Cond.: conditional permutation, only the part of M_k not explained by sector, exposure, exposure^2 and the other nine moderators is permuted; Null m.: null-model Freedman–Lane worlds with residuals sign-flipped by firm.

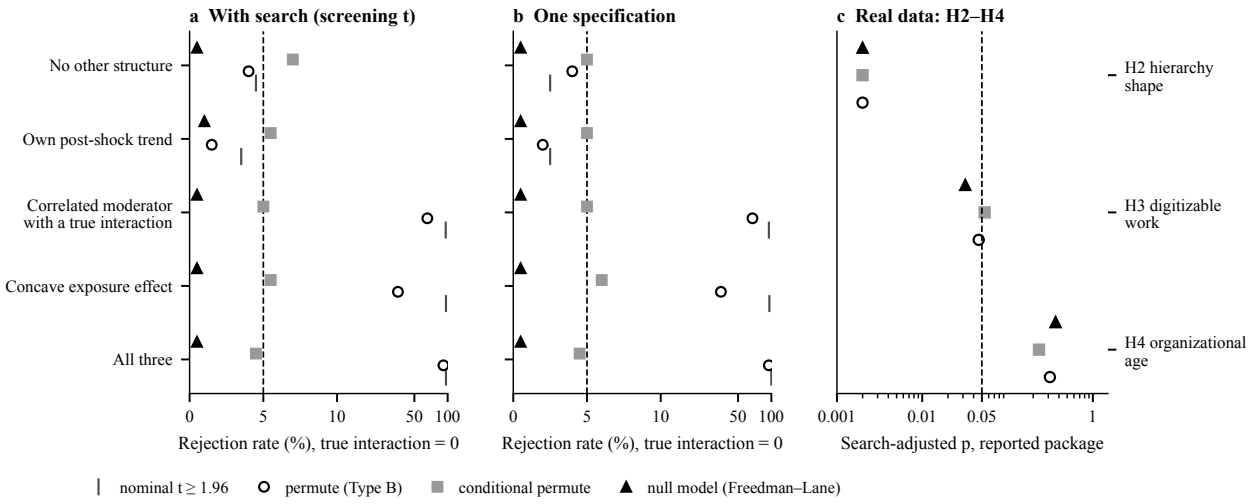


Figure A8: Moderator tests when the true interaction is zero: Monte Carlo rejection rates by construction (a, b), and the real-data p-values of H2–H4 under the three constructions (c).

Table A17: Search-adjusted p-values of the reported moderators, by world construction and statistic

		H2 hierarchy shape	H3 digitizable work	H4 organiz. age
Observed t in the package ($S^{\text{sel}} = S^{\text{fix}}$)		5.57	4.57	2.60
Observed largest screening t (S^{free})		6.27	14.84	6.99
Share of permuted worlds reporting the moderator		0.4%	92.8%	76.6%
Permute (Type B)	S^{sel} (reported package)	0.002	0.046	0.313
	S^{free} (screening, all worlds)	0.002	0.601	0.002
	S^{fix} (reported specification)	0.002	0.018	0.022
	Holm over ten candidates (S^{sel})	0.020	0.413	1.000
Conditional permute	S^{sel} (reported package)	0.002	0.054	0.234
	S^{free} (screening, all worlds)	0.002	0.337	0.004
	S^{fix} (reported specification)	0.002	0.058	0.228
	Holm over ten candidates (S^{sel})	0.020	0.485	1.000
Null model (Freedman–Lane)	S^{sel} (reported package)	0.002	0.032	0.365
	S^{free} (screening, all worlds)	0.002	0.693	0.220
	S^{fix} (reported specification)	0.002	0.184	0.146
	Holm over ten candidates (S^{sel})	0.020	0.287	1.000
Joint (all ten permuted)	max-T step-down (planned), S^{sel}	0.171	0.279	0.651
	min-P step-down, S^{sel}	0.010	0.678	0.678
	min-P step-down, S^{free} (all ten tested)	0.009	0.979	0.009

Notes. $p = (1 + \#\{b : S_b \geq S_{\text{obs}}\})/(B+1)$ with $B = 500$ worlds per moderator and construction ($B = 1,000$ joint worlds); the smallest attainable p is 0.002 (0.001 for joint worlds). S^{sel} is $-\infty$ when the loop does not report the moderator, so its p combines strength and selection. Holm over ten treats the seven unreported candidates as $p = 1$.

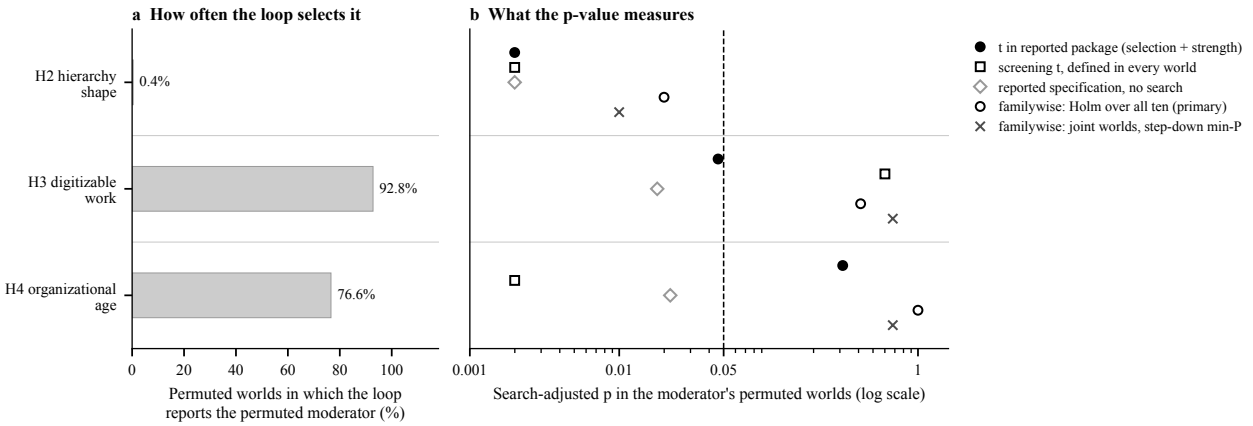


Figure A9: What the moderator *p*-value measures: the share of permuted worlds in which the loop reports the permuted moderator (a), and the *p*-values of the three statistics with the familywise adjustments over ten candidates (b).

8.4 What replay costs

One full replay of the registered loop (both policies, the whole design space, about 5,647 model fits) takes 2.5 seconds with the `nullworld` engine on the real panel. Refitting the same models with `pyfixest` takes about 0.032 seconds each, or 182 seconds per world, roughly 72 times longer. At 1,000 worlds, that is the difference between under an hour and several days (`scripts/27_benchmark.py`).

9 The registry

The registry below was frozen and hashed (SHA-256 prefix `9aec7d8707aa63a2`, 2026-09-29 04:19:18 BST) before any outcome was computed. Deviations after freezing are listed in the design log (`intermediate/00_design_decisions.md`):

- the handling of missing moderators, decided before outcomes were seen;
- the adoption variable, decided before any IV estimate;
- three ex post identification checks;
- a reduced planted-world design, for compute reasons.

The listing omits the file's four-line `authoring` note (lines 11–14); the hash refers to the complete file in the repository.

```
# Registered design space (version 1)
# Frozen BEFORE any outcome regression was estimated. At the time of writing,
# no firm-level outcome (hiring by seniority) had been computed or inspected;
# only data coverage, linkage counts and the exposure crosswalk had been
# examined (see intermediate/00_design_decisions.md). The SHA-256 of this
# file is stored in registry/design_space.sha256 and cited in the manuscript.
# Any change creates version 2 with a dated reason; all versions are reported.

version: 1
frozen_at: "2026-09-29"
```

```
shock:
  event: "Public release of ChatGPT, 2022-11-30"
  post_years: [2023, 2024, 2025]
  pre_years: [2016, 2017, 2018, 2019, 2020, 2021, 2022]
  reference_year_event_study: 2022

unit: firm-year (Compustat NA gvkey linked to Revelio rcid; calendar years 2016-2025)

outcomes: # construct: experience tilt of hiring
  - name: tilt_exp
    definition: "hires with seniority >= 3 / all hires in year t"
    h1_sign: "+"
  - name: hire_sen_mean
    definition: "mean Revelio seniority (1-7) of hires in year t"
    h1_sign: "+"
  - name: entry_rate
    definition: "100 x hires with seniority <= 2 / headcount in year t-1"
    h1_sign: "-"

treatments: # construct: predetermined LLM exposure of the workforce (x Post)
  - name: expo_beta
    definition: "sum_r s_ir(2019-2021) * E_r, E_r = posting-weighted Eloundou et al. (2024) GPT-4 beta"
  - name: expo_human
    definition: "as expo_beta with human-rated beta"
  - name: expo_lm
    definition: "as expo_beta with Felten, Raj & Seamans language-modeling AIOE"
negative_control_treatment:
  - name: expo_nonlang
    definition: "residual of Felten et al. general AIOE on language-modeling AIOE (role level), aggregated as
        above"
    expected: "no post-2022 effect"

h1:
  statement: >
    Firms whose pre-shock workforce was more exposed to large language models shift
    their hiring toward experienced workers after the release of ChatGPT (positive
    effect of exposure x post on the experience tilt of hiring).
  mechanism: >
    Generative AI substitutes for the codified outputs (drafts, code, summaries,
    routine analysis) through which entry-level hires produce value and learn, but
    not for experienced workers' judgment in evaluating and integrating those outputs;
    managers staffing work therefore shift marginal hires up the experience ladder.
  rival: >
    Skill-compression evidence (AI helps novices most) predicts the opposite sign:
    firms could substitute cheaper novices made productive by AI.

moderators: # all predetermined (<= 2021), standardized; sign relative to the H1 effect
  - name: ilm_share
    construct: internal labor market strength
    definition: "share of manager-and-above hires filled by internal movers, 2016-2019"
    source: Revelio positions (earlier window; different construct from outcome)
    sign: attenuates
    mechanism: "Firms that make rather than buy talent depend on entry ports to grow future managers; cutting
        juniors starves their promotion pipeline (Doeringer & Piore 1971; Bidwell & Keller 2014)."
  - name: slack_cash
    construct: financial slack
    definition: "cash and short-term investments / total assets, FY2019-2021 mean"
    source: Compustat
    sign: attenuates
    mechanism: "Slack funds investments with delayed payoffs such as apprenticeship and buffers pressure for
        immediate savings (Cyert & March 1963; Bourgeois 1981; Nohria & Gulati 1996)."
  - name: analysts
    construct: capital-market short-termism
```

```
    definition: "log(1 + distinct analysts issuing FY1 EPS forecasts), 2019-2021 mean"
    source: I/B/E/S
    sign: amplifies
    mechanism: "Analyst scrutiny rewards visible, near-term cost savings, favouring substitution of juniors (
        Graham, Harvey & Rajgopal 2005; He & Tian 2013)."
  - name: firm_age
    construct: organizational age (structural inertia)
    definition: "log(1 + years since founding), founding year from Revelio mapping, else first Compustat year"
    source: Revelio mapping / Compustat
    sign: attenuates
    mechanism: "Older organizations have more institutionalized staffing routines and redesign them more
        slowly (Hannan & Freeman 1984; Sorensen & Stuart 2000)."
  - name: remote
    construct: digitizability of work
    definition: "mean Revelio remote-suitability of active positions, 2019-2021"
    source: Revelio positions (field not used by outcome or treatment)
    sign: amplifies
    mechanism: "Work already codified and performed digitally is easier to reallocate to AI tools."
  - name: ai_talent_pre
    construct: absorptive capacity for AI
    definition: "share of active workforce with core (pre-generative) AI skills, 2019-2021"
    source: Revelio skills
    sign: amplifies
    mechanism: "Prior AI capability lets firms deploy generative AI faster and more deeply (Cohen & Levinthal
        1990)."
  - name: pyramid
    construct: hierarchy shape
    definition: "headcount at seniority 1-2 / headcount at seniority >= 4, 2019-2021 (log)"
    source: Revelio positions (stock, not flow; earlier window)
    sign: amplifies
    mechanism: "Bottom-heavy hierarchies hold more of the codified junior task mass that AI can absorb."
  - name: entry_wage
    construct: labor cost of juniors
    definition: "log mean salary of positions at seniority 1-2, 2019-2021"
    source: Revelio positions (salary field)
    sign: amplifies
    mechanism: "Substitution saves more where juniors are expensive."
  - name: hhi_ind
    construct: product-market concentration (inverse competition)
    definition: "sales Herfindahl index of the firm's NAICS-3 industry, FY2019-2021 (Compustat)"
    source: Compustat
    sign: attenuates
    mechanism: "Concentrated industries face weaker pressure to adopt cost-saving technology quickly."
  - name: rd_int
    construct: knowledge intensity
    definition: "R&D / total assets, FY2019-2021 mean (missing R&D = 0 with indicator)"
    source: Compustat
    sign: attenuates
    mechanism: "Tacit, firm-specific knowledge is transmitted through apprenticeship that AI cannot replace (
        Polanyi 1966; Kogut & Zander 1992)."

controls:
  C0: []
  C1: [log_hc_l1, hire_rate_l1, sep_rate_l1, log_at_l1, roa_l1, lev_l1, sales_growth_l1, rd_at_l1]
  C2: "C1 + (pre-period log headcount) x year dummies + (pre-period hiring growth 2019-2021) x year dummies"

fixed_effects:
  FE1: [firm, year]
  FE2: [firm, naics3_x_year]

samples:
  S1: "2016-2025; firm-years with >= 10 hires"
  S2: "2016-2025; firm-years with >= 30 hires"
```

```
  S3: "S1 restricted to firms observed in >= 8 of the 10 years"

transforms:
  W0: none
  W1: "winsorize outcomes and continuous controls at 1st/99th percentiles by year"
standardization: "treatment and moderators standardized by cross-firm pre-period SD"

clustering:
  CL1: firm
  CL2: naics3

search_policy:
  order: "lexicographic over outcomes x treatments x controls x fixed_effects x samples x transforms x
       clustering, each in the order listed"
  unguarded_U: >
    For each base specification: fit H1; if the sign-consistent t >= 1.96, fit each
    moderator singly (E x Post x M plus M x Post); keep moderators significant with
    the registered sign; try up to 10 triples of mixed sign, ordered by the sum of
    single |t|; success if all four sign-consistent t >= 1.96 in the joint model.
    Stop at the first success; otherwise report the package with maximal
    T = min of the four sign-consistent t.
  unguarded_free_U: "as U but moderator signs unrestricted (ex post rationalization)"
  package_statistic: "T = min over H1 and the three moderators of sign-consistent t"
  h1_statistic: "max over visited base specifications of the sign-consistent t for H1"

guard:
  split: "firms randomly assigned 50/50 to exploration/confirmation within NAICS-2 strata"
  seed: 20261101
  confirmation_model: "the package selected on exploration, re-estimated once on confirmation"
  multiplicity: "Holm across H1-H4 at alpha = 0.05"

null_worlds:
  typeA: {permute: treatment (firm-level), within: naics2, B: 1000}
  typeB: {permute: all moderator candidates independently, within: naics2 x treatment_quintile, B: 500}
  seeds: "world b uses seed 1000003 * b + 17"
planted_worlds:
  beta_star_within_sd: [0.02, 0.05, 0.10]
  gamma_star_within_sd: [0.02, 0.05]
  worlds_per_cell: 200

identification_checks: [event_study_pretrends, placebo_2019, placebo_2020, negative_control_exposure,
  naics3_x_year_fe, drop_information_sector, pre_hiring_growth_x_year, remote_x_year,
  rotemberg_top_roles_dropped, honest_did_relative_magnitudes, 2sls_genai_share, randomization_inference]
```

10 Repository and reproduction

The research artifact submitted with this appendix is the public repository `https://github.com/LLMPlus-ai/potemkin-findings`. Its numbered scripts write every number, table, and figure, and each records its inputs and outputs in its header.

- `registry/` holds the frozen design space and its SHA-256 hash; every run of the loop verifies the hash first.
- `nullworld/` is the reusable package: fixed-effects engine, replayable loop, world generators, randomization inference, synthetic twin, and registry grammar (Section 2).
- `scripts/01`–`07` build the data from the licensed sources (DuckDB, memory capped at 4 GB); `scripts/08` fits and releases the synthetic twin.

- `scripts/09–14` estimate the testbed, run the loop on the real data, generate the null, guard, and planted worlds, run the Monte Carlo study, and aggregate Potemkin rates and search-adjusted p-values.
- `scripts/22–27` produce the validity diagnostics of Section 8 and the cost benchmark.
- `scripts/12`, `15`, `16`, `19–21`, and `26` write the figures and tables; `scripts/17` and `25` write every number cited in the text as a macro.
- `scripts/18` verifies every reference against Crossref (`literature/citation_audit.csv`); `scripts/28` recomputes the cited validity numbers independently from the stored worlds; `scripts/90–92` validate the engine against `pyfixest`.
- `intermediate/` stores the output of every world (Type A, Type B, joint, trait-preserving, conditional, rebuilt-outcome, guard, and planted) and every Monte Carlo replication.

Without licensed data. The Revelio, Compustat, and I/B/E/S extracts cannot be redistributed and are not in the repository. With `NULLWORLD_DATA=synthetic`, the loop and all null worlds (`scripts/10–11`) run on the synthetic twin in `synthetic/`. The Monte Carlo studies (`scripts/13`, `23`, `24`) simulate their own twins, and the figure and table scripts read only the stored outputs. The scripts that read the licensed data directly (01–09, 12, 14, 22, 25, 27; script 08 to fit the twin) document every field they use, so licensees can regenerate every number.

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